

Po-Chun Wu

pcwu2022@gmail.com | (412) 906-0182 | [linkedin.com/in/pcwu2022](https://www.linkedin.com/in/pcwu2022)

Education

Carnegie Mellon University

Master of Science in Electrical and Computer Engineering

Master of Science in Engineering and Technology Innovation Management (Dual Degree)

Pittsburgh, PA

May 2028

May 2028

National Taiwan University

Bachelor of Science in Electrical Engineering

GPA: 4.16/4.3

Taipei, Taiwan

Jan 2026

Relevant Courses: Introduction to EDA, SoC Verification, Data Structures, Algorithms, Computer Architecture, Digital IC Lab

Projects

Obamify – NTU, Course Project: Digital Circuit Lab (Team of 3)

Nov 2025 — Dec 2025

- Designed a pixel-wise image transformation algorithm using SystemVerilog, programmed on FPGA boards.
- Developed a Python architectural reference model to simulate memory constraints, FSM cycles, and bitwise operations.

Reverse Engineering – NTU, Course Project: Introduction to EDA (Team of 3)

Nov 2025 — Dec 2025

- Reconstructed arithmetic RTL code from gate-level Verilog using BDD and *BMD data structures for small instances.
- Used large language models to identify circuit structures and input-output relationships.

Formal Verification – NTU, Research Project (Advising: Prof. Chung-Yang Ric Huang, Team of 2)

Apr 2025 — Jun 2025

- Modeled optimal vocabulary learning sequence generation as an NP-hard Target Set Selection variant.
- Implemented bounded model checking-based verification to derive optimal solutions for graphs with 150 nodes.

LifeOS Automation System – NTU, Entrepreneurship Program (Team of 6)

Jan 2025 — Mar 2025

- Engineered a timestamp-based version-state model to resolve sync conflicts between Notion pages and MongoDB records.
- Developed a document-linking layer mapping a canonical schedule to multiple nodes for one-to-many synchronization.

CPU Design – NTU, Course Project: Computer Architecture (Team of 2)

Apr 2023 — Jul 2023

- Designed a multi-cycle CPU running on RISC-V instructions using Verilog, ranking in the top 15% of the class.
- Added a 32-block direct-mapped cache to reduce latency, achieving a 50% improvement in speed.

Work Experience

SLEKMED

Taipei, Taiwan (Remote)

Product Developer (Part-Time)

Aug 2023 – Mar 2026

- Note-to-Video Generator:** Developed a web-based note-to-video generator using Marp, OpenAI API, ElevenLabs API, and FFmpeg, enabling medical experts to create high-quality educational videos from Markdown notes in less than 10 minutes.
- Medical Simulation Platform:** Created a Next.js-based simulation module for medical diagnostic training in collaboration with Taipei Medical University peers, gaining 4,200+ followers and securing funding from the Taiwan Ministry of Education.
- Virtual Patient Chatbot:** Implemented an LLM-based chatbot as a virtual patient, contributing to a 1st-place finish among 40+ teams in the 2024 CTCI AI Competition (12K USD prize).

Skills

Programming: Verilog, C++, Python, JavaScript/TypeScript

Frameworks/Tools: MiniSat, FPGA, Quartus, BDD, React/Next.js, MongoDB, Git

Specialties: System design, algorithm design, formal verification, RTL development, full-stack development

Leadership

Founder and President, NTU Learning Optimization Club

Aug 2023 – Jun 2025

- Researched learning methodologies, delivered 8 nationwide speeches, and advised 200+ students on study habits.

Makerspace Administrator, NTUEE Student Association Academic Department

Sep 2023 – Dec 2023

- Organized 25 academic workshops and taught microcontroller programming to electrical engineering undergraduates.

Project Lead, NTUEE Virtual Makerspace

May 2023 – Aug 2023

- Led a 25-member team to build a React/Node.js inventory and reservation system for the university student makerspace.